

Glaucoma

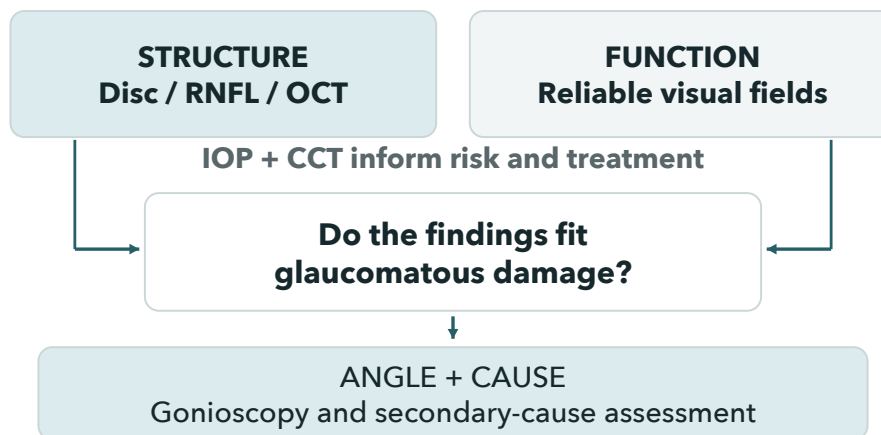
M3-M5 | Confirm damage and mechanism, individualise pressure lowering

Confirm the disease, then the mechanism

Clinical definition

Glaucoma is a progressive optic neuropathy with characteristic structural damage; visual field loss may follow. Pressure is a modifiable risk factor: a normal reading does not exclude glaucoma, and elevated pressure alone does not establish it.

Three questions at diagnosis



Original assessment map | Combine structure, visual function and angle anatomy; interpret pressure in context.

Terms with different consequences

Finding	Interpretation
Raised IOP; nerve and fields normal	Ocular hypertension. Risk-based review and treatment.
Open angle + glaucomatous damage	Open-angle glaucoma. Field loss may appear later.
Typical damage; usual-range IOP	Normal-tension glaucoma; exclude mimics.
Angle closure ± high IOP / synechiae	Stage as suspect, closure or angle-closure glaucoma.

Baseline assessment

Goldmann applanation, gonioscopy, disc / retinal nerve fibre layer assessment, OCT, reliable automated fields and central corneal thickness. Check steroid use, pseudoexfoliation, pigment dispersion, trauma and iris neovascularisation. Interpret disc measurements and OCT colour coding in context.

References Singapore National Eye Centre | Glaucoma | AAO EyeWiki | Ocular hypertension | NICE NG81 | Recommendations, updated 2022

POAG and ocular hypertension: initial plan

Case discussion

A patient has an open angle, reproducible early field loss and matching rim thinning. Arthritis makes drops difficult. Choose pressure lowering that fits the patient's capabilities, goals and follow-up.

NICE NG81 initial treatment pathway

- 1** New nonadvanced open-angle glaucoma: offer 360° selective laser trabeculoplasty (SLT). Exclude pigment dispersion.
- 2** Explain that SLT can reduce or delay drops; it may wear off, require repeat treatment or still need medication.
- 3** If SLT is declined, unsuitable, insufficient, or needed treatment cannot wait: offer a generic prostaglandin analogue.
- 4** Advanced disease: discuss glaucoma surgery and interim pressure lowering. Escalate according to progression and target pressure.

Risk and treatment goals

Issue	Clinical use
Ocular hypertension	Treat if lifetime visual impairment risk warrants it; pressure alone is insufficient.
Damage / rate of progression	Set an individual target IOP; advanced or rapidly progressing disease needs greater protection.
Age, life expectancy, CCT, family history	Help estimate risk and follow-up needs.
Physical ability, adherence, ocular surface	Influence the choice between laser, drops and surgery.

Local context

The SLT-first offer follows UK NICE guidance. Singapore care is individualised by the ophthalmologist. Pressure lowering preserves remaining vision; ongoing review continues after medication, laser or surgery.

Pressure-lowering drugs: choose safely

Mechanism plus the patient who changes the choice

Class / example	Action	Important checks
Prostaglandin analogue Latanoprost	Increases outflow	Hyperemia, lash / iris changes; assess inflammation and other ocular risks.
Beta blocker Timolol	Reduces aqueous production	Avoid in asthma, sinus bradycardia and significant heart block; topical drops have systemic effects.
Topical carbonic anhydrase inhibitor Dorzolamide	Reduces aqueous production	Local stinging; assess renal impairment and relevant hypersensitivity.
Alpha-2 agonist Brimonidine	Reduces production; increases outflow	Allergic conjunctivitis, fatigue / hypotension; age and cardiovascular cautions.
Systemic CA inhibitor Acetazolamide	Rapidly reduces aqueous production	Check renal / liver disease, electrolytes, acidosis and relevant drug reactions.
Miotic Pilocarpine	Contracts ciliary muscle and pupil	Mechanism-specific use; not the default for all angle closure.

If pressure remains above target

- 1 Check instillation, adherence, tolerability, supply and interacting medicines.
- 2 Repeat relevant pressure / field / structural assessment; distinguish poor delivery from true progression.
- 3 Optimise or add a different class, laser or surgery with the glaucoma team.

Teach the technique

Instil one drop; avoid bottle-tip contact. Close the lids and use punctal occlusion to reduce systemic absorption. Space different drops. Check combination-drop ingredients: some contain timolol.

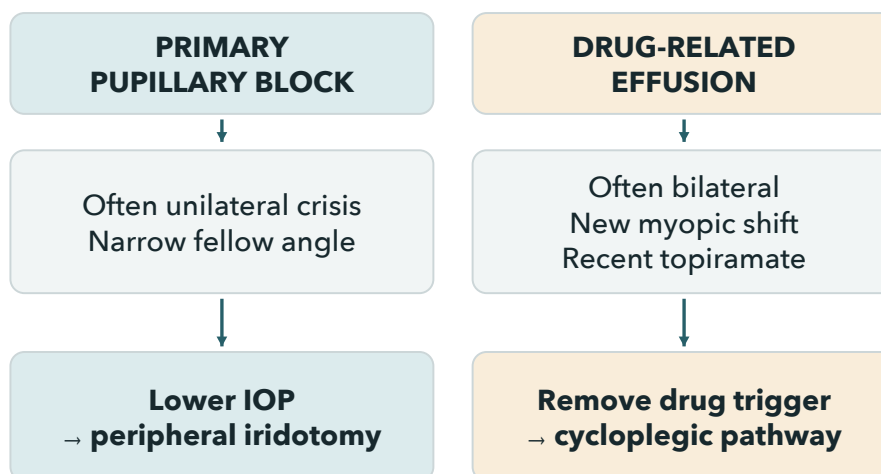
References AAO EyeWiki | Topical glaucoma medicines | DailyMed | Timolol ophthalmic prescribing information | DailyMed | Acetazolamide prescribing information

Acute angle closure: recognise and act

Case pivot

A painful red eye with haloes, nausea, a hazy cornea, shallow chamber, mid-dilated poorly reactive pupil and markedly raised IOP is an emergency. Determine whether pupillary block or a secondary mechanism is responsible.

The mechanism changes treatment



Original decision schematic | Laser peripheral iridotomy treats pupillary block; secondary causes need their own pathway.

Typical primary pupillary-block crisis

- 1 Immediate ophthalmic care; analgesia / antiemetic and pressure lowering with appropriate topical agents and systemic acetazolamide after contraindication checks.
- 2 At very high IOP, defer pilocarpine until pressure falls and iris responsiveness improves; avoid intensive repeated dosing.
- 3 Laser peripheral iridotomy as soon as technically possible; assess and protect the at-risk fellow eye.
- 4 Reassess angle, pressure and optic nerve. Lens extraction may be appropriate in selected angle-closure disease / cataract.

Recognise secondary mechanisms

Bilateral shallow chambers and sudden myopic shift after topiramate suggest ciliochoroidal effusion. Stop the drug with the prescribing team; obtain urgent specialist care for cycloplegia and directed treatment. Pilocarpine can worsen this mechanism; iridotomy does not correct it.

Secondary causes and follow-up decisions

Recognise what pressure treatment alone misses

Clinical clue	Further action
Steroid exposure + pressure rise	Review the steroid indication and adjust under supervision; monitor pressure and nerve.
Rubeosis / angle neovascularisation	Find retinal ischaemia; treat the retinal driver alongside pressure lowering.
Trauma; asymmetric angle findings	Look for angle recession or other injury; long-term glaucoma surveillance may be needed.
Pseudoexfoliation / pigment signs	Identify the subtype; pressure behaviour and treatment suitability can differ.

At each glaucoma review

- 1 Ask about symptoms, daily treatment, adverse effects and functional difficulties; observe drop use.
- 2 Compare IOP with the individual target, not a generic normal range.
- 3 Assess reliable fields and optic nerve / OCT trends. Confirm apparent progression and investigate discordance.
- 4 If progressing, revise target and treatment. If stable, continue risk-based review rather than declaring a cure.

Counselling that matters

Explain that good central acuity can coexist with substantial field loss and that treatment does not restore established damage. Laser does not guarantee permanent freedom from drops. New pain, redness or sudden visual decline warrants urgent reassessment rather than waiting for routine glaucoma follow-up.

Reference map

Initial offers: NICE NG81. Local context: SNEC. Drugs: AAO review and prescribing information. Acute angle closure: RCOphth. Secondary mechanisms: AAO. Original teaching schematics.